Random Variables & Distributions Transcript

Hi everyone, let's get into the second module that or the second lecture that you have on random variables and different types of random variables. We will get introduced to random variables, what it is, why we use it and what different types of random variables do we have and so on. But let us first start with a simple analogy, we might have been traveling from point A to point B many number of times, let's say you might have been traveling to airports from your home, sometimes it might have been taking 500 rupees, sometimes it might have been taking 700 or 1000 whatever.

But have you noticed the variance which comes in, the number of times that you travel to airport, the amount that you pay to taxis or Ola or Uber or whatever mechanism that you have via bus or train, your amount is not the same every time, right, that's what is a variable. Variable as in something which varies. Now a random variable is which assumes any number randomly.

Now think of in between the total amount of time taxis might be charging, it might be in between 0 to 20000, right, and whenever you travel to airport, you might have to pay either 5000 or 10000 or 1000 whatever it is, but it will take that amount within that fixed interval. So that's what is called as random variable and a random variable could be of different types and that's what we are going to learn in this and let's get started. So random variables, we define these as something which happens randomly, right, so think of coin tosses, you have only two outputs which are head and tail, when we assign a number to random variables or the outputs of random variables, we call that as a random variable and the outputs are numeric, so the moment we assign numbers to anything which is random, we call all of these things as random variables.

Different types of things can happen, think of how many different types of numbers that you can attach. Let's say you are producing a video, right, a video content might be good or bad, the video that you are recording or the image that you are capturing, it might be good or bad, so good can be represented as 1, bad can be represented as 0, so you are assigning a number, a numeric number to a random variable which is clicking an image or shooting a video, right, the random experiment is clicking an image or shooting a video, but the output could be 1 or 0, so the moment you assign a value, it becomes variable and the variable can vary because sometimes it would be 0 which means the output is bad and sometimes it could be 1, the output could be really really good, right, and think of these are two different outputs, either you would be 0 or 1 in this case, but when you travel from airport to your home or your home to airport, you might sometimes pay very less, 200 rupees, sometimes pay 2000 rupees, sometimes even pay higher, right, that itself is a range of values and I am giving you a number which still cannot be in decimals, think of different types of heights people have, I might be 5'6, someone might be 5'7, some other can be 5'8, those values are continuous, someone's height cannot be measured exactly with 5'6, right, so you cannot say that I am exactly 5'6 or 5'8, 6 inches tall, similarly when you measure any length, length cannot be specific, it cannot be certain, it cannot be a number which is always a specific number, right, so we call these things as continuous, so in totality a random variable can have two things or two outcomes or random variables can be defined in two different categories, one discrete, the example, remember the example that I gave, right, a photo that you have clicked from your phone, it could be good or bad, so it has two specific outputs and you have attached numbers to it, 0 and 1, so you have a specific output, that's why they are called as discrete, when you have various number of outputs in a range, let's say height, weight, think of any physical parameter that you have, think of time as well, right, when you notice time, it always flows, you can divide a time from hours to minutes to seconds to milliseconds to whatever, right, you cannot pinpoint a specific thing in time, that's what is called as continuous random variable, random variable again if I have to tell you again, think of anything which varies and assumes a number randomly, when the process of assigning a number to something which happens randomly is called as the random variable and random variable can be of two different types, one is discrete and continuous, think of a company which produces different types of products, right, products could be good or bad or it can, the number of products can never be 2.5, the number of products could be 100, think of Britannia producing biscuits, they could produce 100s of biscuits, they cannot produce 100.5 amount of biscuits or 1000.25 amount of biscuits, that is a specific number, they will never be producing something which is in decimal, that's why we call them as discrete, discrete as in something which is a specific and continuous as something which has a very smooth range, think of height, weight, temperature, pressure, any other quantity, right, so the example that you are seeing on your screen right now, think of a ball falling from a particular height and going into different types of wind, every time you do that experiment, you might see in this case, the ball falling, the GI that you could see on your screen, right, it might fall in different bins every time that you visualize it, so that is a random process, right, and the random process if you can think of, if you could let me highlight it, let's say this is number 1, this is bin number 2, this is bin number 3 and so on, now this ball can take the entirely new route and go to number 1 and it could also go to number 2 or it could go to number 3 or bin 3, whatever, right, so there could be various different types of cases which can happen, this is very much relatable to the example that I gave you, for you travelling from your home to airport, let's say for example, this is your home and this is airport, right, and this is where you might be getting charged 100 rupees or this is where you might be getting charged 1000 rupees and so on, so things are quite relatable, it's just that we will have to sometimes relate to it in much more real-life examples, that's what I am trying to help you with, so this is what is called as random variable, now on your screen you are able to see that when you roll a dice, right, or let's say a non-bias dice which lands evenly on any of those outputs, a dice could have 6 faces which has 1, 2, 3, 4, 5, 6, when you roll a dice you could get either of these numbers, right, so when we roll a dice you might get 1 in 1 first outcome, you might also get 1 again and there could be various different types of outcomes that you can get in every random throw, now the process itself if you think about, this is a random experiment, right, because you are throwing a dice, dice throw itself is a random experiment, a random variable is what is the outcome every output of that dice throw has, so let's say dice throw would either assume 1, 2, 3, 4, 5, 6, but if I attach probabilities to these numbers, let's say the probability of getting 1 is equals to 1 by 3 because every particular number is equally likely, right, so the probability of every number could be 1 by 6 and that's what it is because there are 6 different types of numbers which are there on your dice and if you land to one specific number as an outcome when you roll a dice, that probability itself would be 1 by 6 and that's what we are visualizing here when we roll dice and what happens, so on the right hand side you see that there these 6 cases, right, and while in the second case think of the heights of people, okay, which are arranged in a continuous fashion, if I go back to my board and which wherein I could help you understand this, think of a random variable as two different types of things, one is discrete, the other one is continuous, in discrete cases the event would be specific, the output would be specific, in continuous cases they will always be a continuous value, okay, and when you roll a dice it's discrete because you only have a fixed number of outcomes, 1, 2, 3, 4, 5 and 6, let's say this is 1, this is 2, this is 3, this is 4, this is 5 and this is 6, what is the probability of getting 4 when you throw a dice, it's exactly 1 by 6 because you have only one favorable outcome which is 4 but total number of outcomes is in total 6, right, when in denominator you have total number of outcomes that could happen, right, in your sample space and in your numerator you have what you want as an outcome, right, what which is favorable, the probability in this case is favorable divided by outcomes hence it would be 1 by 6, similarly let's say for example if I have to find out the probability that someone's height is greater than 5, so in this particular case let's say the height could be ranging from 2 feet to let's say 8 feet, right, now if I have to find out because every individual's height would be a continuous number, it could be ranging from 2 to 2, 2 to 3, 3 to 4 and then so on, right, let's say these are all your intervals and that's what you have plotted, now every single range has been plotted here, now if you would want to know in this case what is the probability that someone's height is greater than let's say 4, so this is that shaded region, now why I have plotted a continuous curve because someone's height could be exactly 5.6 and we cannot estimate it as exact 5.6, right, we would always have something like 5.6 or 5.602 and so on, right, it cannot be as exact as 5.6 that's what we say, so that's what is continuous, continuous can never attain a fixed value, so I just wanted to help you introduce with this concept which is random variable, I hope this explanation and everything that I gave you was enough for all of you to understand it, okay, let's meet in the next class, thank you.

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